



## Introduction

# Motivation and Context

The Cognitive Integrity Framework (CIF) introduced in Part 1 of this series establishes formal foundations for securing multiagent AI operators against cognitive manipulation attacks. This companion paper provides comprehensive empirical validation, demonstrating that CIF's theoretical constructs translate into practical, deployable protection mechanisms.

## The Theory-Practice Gap

Formal security guarantees, while essential for theoretical confidence, face a critical question: *do they work in practice?* The history of security research is replete with mechanisms that succeed in controlled settings but fail when confronting real adversaries, production workloads, and architectural constraints. The gap between theoretical security and practical deployment arises from several factors:

- ▶ **Adversarial adaptation:** Real attackers probe defenses and evolve tactics; theoretical bounds assume fixed attack distributions
- ▶ **Implementation fidelity:** Production systems introduce

# Paper Contributions

## COGNITIVE INTEGRITY FRAMEWORK (CIF)

Layered Defense Architecture for Multiagent AI Systems



Figure 1: CIF Comprehensive Architecture. Overview of the Cognitive Integrity Framework showing the relationships between the five core

# Relationship to Paper Series

This paper assumes familiarity with the formal framework developed in Part 1, particularly:

- ▶ **Trust Calculus** (Section 3 (Trust Calculus, Part 1)): Bounded delegation with  $\delta^d$  decay
- ▶ **Defense Composition Algebra** (Section 4 (Defense Composition, Part 1)): Series and parallel composition theorems
- ▶ **Integrity Properties** (Section 5 (Integrity Properties, Part 1)): Belief consistency, goal preservation, trust boundedness

All notation follows the canonical reference in Part 1 Appendix (sec:notation-reference). For practical deployment guidance including checklists and operational considerations, see Part 3.

# Paper Organization

The remainder of this paper is structured as follows:

**sec:methodology: Methodology** presents implementation details for each defense mechanism.

**sec:attack-corpus: Attack Corpus** describes the 950-attack evaluation dataset with examples and generation methodology.

**sec:experimental-setup: Experimental Setup** details the six target architectures and evaluation protocol.

**sec:results: Results** presents detection performance, ablation studies, and scalability analysis.

**sec:analysis: Analysis** provides statistical significance testing and cross-architecture comparison.

**sec:discussion: Discussion** examines limitations, deployment considerations, and future work.

**sec:conclusion: Conclusion** summarizes contributions and identifies next steps